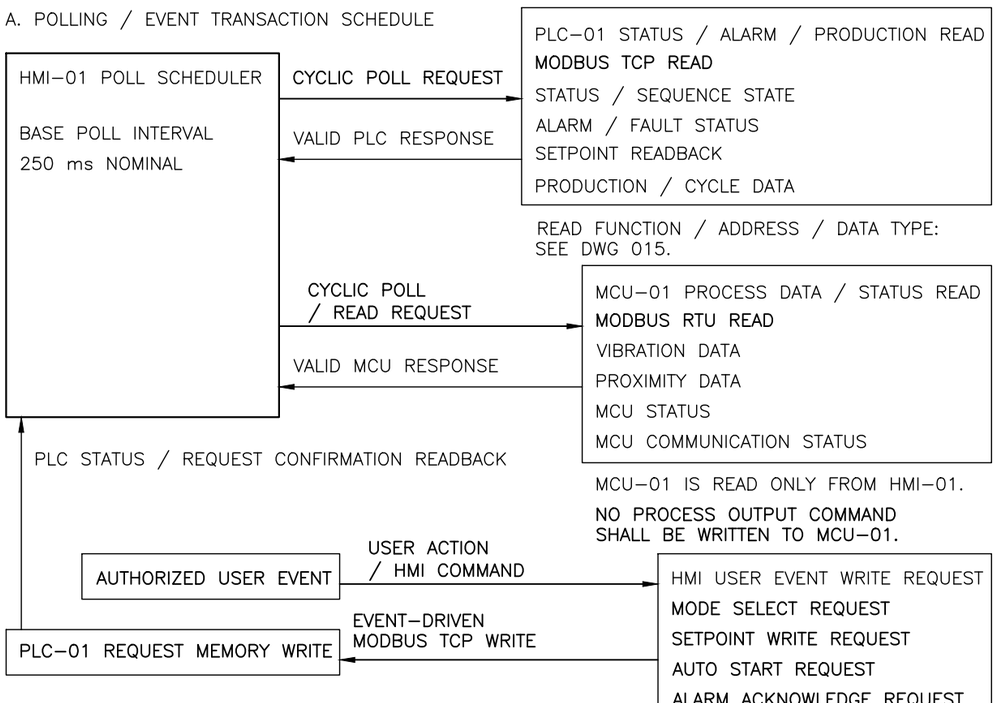
HMI–01 MODBUS TRANSACTION / POLLING / COMMUNICATION FAULT LOGIC

PLC–01 MODBUS TCP / MCU–01 MODBUS RTU / HMI–01 SUPERVISORY SCADA

A. POLLING / EVENT TRANSACTION SCHEDULE

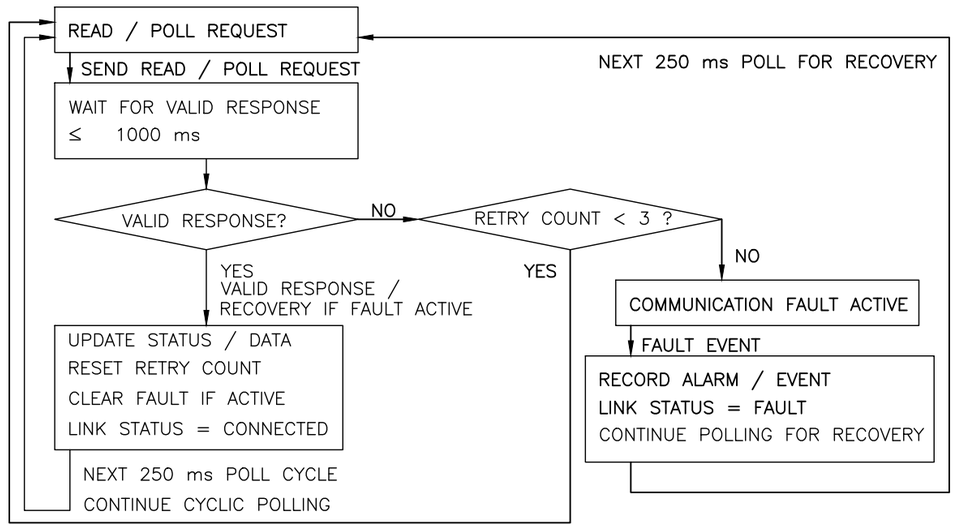


CYCLIC READ POLLING SHALL BE SEPARATE FROM EVENT–DRIVEN COMMAND / SETPOINT WRITE TRANSACTIONS.
HMI–01 SHALL NOT AUTOMATICALLY REISSUE A COMMAND WRITE AFTER A WRITE TIMEOUT.

D. COMMUNICATION FAULT / RETRY / RECOVERY FLOW

APPLIES TO:
PLC–01 MODBUS TCP READ / POLL
MCU–01 MODBUS RTU READ / POLL

PLC–01 MODBUS TCP AND MCU–01 MODBUS RTU SHALL USE THE SAME READ / POLL RETRY LIMIT: 3 ATTEMPTS MAXIMUM.



INCREMENT RETRY COUNT
RETRY READ / POLL REQUEST

COMMAND OR SETPOINT WRITE TIMEOUT:
DO NOT AUTOMATICALLY REISSUE WRITE COMMAND.
COMMUNICATION FAULT SHALL NOT BYPASS, DISABLE, OR REPLACE HARDWIRED SAFETY FUNCTION.

B. PLC–01 MODBUS TCP TRANSACTION TABLE

PLC-01 MODBUS TCP TRANSACTION					
No.	Direction	Transaction type	Data group	Trigger / interval	HMI action / result
1	HMI-01 → PLC-01	Read	PLC Status / Sequence State	Cyclic, 250 ms nominal	Update Main / Auto / Diagnostic display
2	HMI-01 → PLC-01	Read	Alarm / Fault Status	Cyclic, 250 ms nominal	Update Active Alarm / History display
3	HMI-01 → PLC-01	Read	Setpoint Readback / Validation State	Cyclic, 250 ms nominal	Update Manual / Setup display
4	HMI-01 → PLC-01	Read	Production / Cycle Data	Cyclic, 250 ms nominal	Update Trend / Production display
5	HMI-01 → PLC-01	Write	Authorized HMI Command Request	User event only	PLC validates request before execution
6	HMI-01 → PLC-01	Write	Authorized Setpoint Request	User commit only	PLC range/mode/state validation required
7	PLC-01 → HMI-01	Response	Request Confirmation / Updated Status	Next valid read cycle	HMI updates command outcome display

MODBUS FUNCTION CODE, REGISTER / BIT ADDRESS, DATA TYPE, AND SCALING SHALL BE DEFINED ONLY IN DWG 015.

PLC–01 SHALL NOT ACCEPT A DIRECT HMI WRITE TO PHYSICAL OUTPUTS P20~P29.

C. MCU–01 MODBUS RTU TRANSACTION TABLE

MCU-01 MODBUS RTU TRANSACTION					
No.	Direction	Transaction type	Data group	Trigger / interval	HMI action / result
1	HMI-01 → MCU-01	Read	Vibration / Process Value	Cyclic, 250 ms nominal	Update Trend / Process display
2	HMI-01 → MCU-01	Read	Proximity Value	Cyclic, 250 ms nominal	Update Trend / Manual / Setup display
3	HMI-01 → MCU-01	Read	MCU Status / Device Health	Cyclic, 250 ms nominal	Update Communication / Diagnostic display
4	MCU-01 → HMI-01	Response	Process Data / Status Response	Valid RTU response	Store Trend / Status data
5	N/A	MCU Write	Not permitted	N/A	No direct HMI process output command

MCU–01 MODBUS RTU FUNCTION CODE, REGISTER ADDRESS, DATA TYPE, AND SCALING: SEE DWG 015.

MCU–01 SHALL PROVIDE PROCESS DATA ONLY.
PROCESS CONTROL AUTHORITY SHALL REMAIN WITH PLC–01.

[TRANSACTION RULES / CROSS REFERENCE NOTES]

1. HMI–01 SHALL INITIATE ALL MODBUS TCP AND MODBUS RTU TRANSACTIONS.
2. PLC–01 SHALL RESPOND AS MODBUS TCP SERVER. MCU–01 SHALL RESPOND AS MODBUS RTU SLAVE.
3. CYCLIC READ POLLING INTERVAL SHALL BE 250 ms NOMINAL.
4. RESPONSE TIMEOUT SHALL BE 1000 ms. MAXIMUM AUTOMATIC READ / POLL RETRY COUNT SHALL BE 3.
5. HMI COMMAND AND SETPOINT WRITES SHALL BE EVENT–DRIVEN, NOT CYCLIC POLLING TRANSACTIONS.
6. HMI–01 SHALL NOT AUTOMATICALLY REISSUE COMMAND OR SETPOINT WRITE AFTER A WRITE TIMEOUT.
7. PLC–01 SHALL VALIDATE MODE, STATE, RANGE, FAULT, AND SAFETY CONDITIONS BEFORE COMMAND EXECUTION.
8. HMI–01 SHALL NOT DIRECTLY WRITE PHYSICAL PLC OUTPUTS P20~P29 OR BYPASS HARDWIRED SAFETY FUNCTION.
9. MCU–01 SHALL PROVIDE PROCESS DATA ONLY. NO DIRECT HMI PROCESS OUTPUT COMMAND SHALL BE PERMITTED.
10. MODBUS FUNCTION CODE, REGISTER / BIT ADDRESS, DATA TYPE, DATA SCALING, AND TAG NAME: SEE DWG 015.
11. HMI SCREEN DISPLAY / ALARM / HISTORY / TREND DESIGN: SEE DWG 016.
12. HMI FUNCTIONAL / DATA LOGGING ARCHITECTURE: SEE DWG 017.
13. PHYSICAL ETHERNET / RS–485 INTERFACE / TIMEOUT SETTINGS: SEE DWG 018 / SHEET 1 OF 2.
14. MODBUS EXCEPTION RESPONSE SHALL BE TREATED AS A VALID COMMUNICATION RESPONSE WITH APPLICATION ERROR, NOT AS A COMMUNICATION TIMEOUT.
    MODBUS EXCEPTION SHALL BE RECORDED AS HMI COMMUNICATION / CONFIGURATION DIAGNOSTIC EVENT.

[TRANSACTION LEGEND]
DIAMOND
DECISION / RESPONSE VALIDATION CONDITION
SOLID ARROWHEAD
TRANSACTION FLOW DIRECTION

[TRANSACTION LEGEND]
YELLOW ARROW
REQUEST / POLL / WRITE TRANSACTION
GREEN ARROW
VALID RESPONSE / STATUS / PROCESS DATA
RED ARROW / BLOCK
FAULT / ALARM / WRITE TIMEOUT RESTRICTION

NO.	DATE	REVISION
A	2026–09–22	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi–Bin Sorting Cell
SCADA MODBUS TRANSACTION / POLLING / COMMUNICATION FAULT DIAGRAM

DRAWN BY: J. H. KO
DATE: 2026–09–22
DWG NO: 018
SHEET NO: 2 OF 2